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Professor Gianfranco Chicco
Editor-in-Chief, *Sustainable Energy, Grids and Networks* (SEGAN)
Elsevier

Subject: Submission of Original Research Article to *Sustainable Energy, Grids and Networks*

Dear Professor Chicco and Editorial Board,

I am pleased to submit my original research manuscript titled:

"Communication-Aware Event-Triggered Primary Frequency and Synthetic Inertia Support from Data Center UPS Fleets"

for consideration as a Research Article in *Sustainable Energy, Grids and Networks* (SEGAN).

Alignment with SEGAN Aims and Scope

This manuscript directly advances several core focus areas specified in the SEGAN Aims and Scope, namely:

1. **Control and planning of centralized and distributed energy storage**: Decentralized allocation of fast power reserves across multi-agent battery arrays while strictly respecting local state-of-charge boundaries and mission-critical backup runtime floors.
2. **ICT integration in electrical and energy systems for monitoring and control**: Communication-aware event-triggering over local area networks with imperfect channels subject to packet drop, log-normal jitter, and variable sensing latencies.
3. **Modeling of cyber-physical energy and communication systems**: Unified frequency-domain and time-domain stability formulations coupling physical power-frequency swing dynamics with discrete network protocols.
4. **Demand-side management and demand response: definition of control schemes and protocols**: Harnessing pre-capitalized computing infrastructure to provide clean grid services without requiring dedicated utility-scale land footprint.

Key Methodological and Practical Contributions

The paper addresses the challenge of declining synchronous inertia in low-carbon grids by leveraging uninterruptible power supply (UPS) batteries in hyperscale data centers. Crucially,

rather than presenting a case study or converter topology design, the paper provides fundamental methodological and computational contributions:

- **Analytical Tracking and Nadir Degradation Bound**: We derive a closed-form analytical theorem (Section 2.3.1) proving that the worst-case frequency nadir degradation under adaptive event-triggering, network delay, and packet loss is strictly bounded by $|\Delta f_{\text{err}}| \leq 8.0\,\text{mHz}$ relative to continuous droop.
- **Communication-Aware Adaptive Thresholding**: We formulate PULSE (Primary-frequency UPS Local Sub-second Event-triggered control), which eliminates idle-state communication under ambient drift and cuts post-contingency message traffic by 63.6% compared to fixed-threshold schemes.
- **Analytical Delay Margin and Stability Proof**: Frequency-domain Nyquist and Bode stability analyses establish an open-loop phase margin of 68.2° and an analytical delay margin $\tau_{\text{margin}} = 1.58\,\text{s}$, exceeding actual network delays by more than twenty-fold.
- **Sensing Realism and RoCoF Filtering**: We reframe sub-cycle synthetic inertia by leading with realistic 1-cycle PLL ($65.0\,\text{ms}$) and 2-cycle PMU ($90.5\,\text{ms}$) closed-loop power rise times, avoiding unphysical derivative chattering and inverter saturation.
- **Battery Autonomy and Degradation Assurance**: Dynamic battery modeling proves that providing 30 s grid support consumes only $0.166\,\text{kWh}$ per rack, maintaining $> 40\,\text{min}$ of autonomous IT backup runtime (100% Tier III/IV compliance) with negligible 10-year cycling wear ($< 0.018\%$).

Contribution to UN Sustainable Development Goals (SDGs)

This work directly supports **UN SDG 7 (Affordable and Clean Energy)** and **SDG 11 (Sustainable Cities and Communities)** by transforming massive, pre-capitalized, and otherwise idle data center energy storage into low-carbon cyber-physical grid stability reserves, lowering reliance on fossil-fuel peaker plants.

Declarations and Submission Details

- **Sole Authorship**: The author is the sole contributor and conducted all conceptualization, modeling, simulation, formal analysis, and manuscript preparation.
- **Originality**: This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere.
- **Conflict of Interest**: The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence this work.
- **Data Availability**: All simulation code, automated verification suites, and numerical datasets are openly accessible via GitHub and permanently archived on Zenodo with DOI: 10.5281/zenodo.14920188.
- **Generative AI Statement**: AI assisted with language editing and grammar polishing. The author reviewed and edited all content and takes full responsibility for the published article.

Thank you very much for your time and consideration of this manuscript.

Sincerely,

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